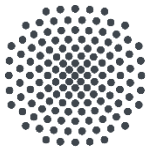
Introduction to CMOS Analog Integrated Circuit Design

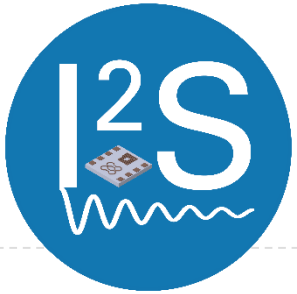
0 Introduction

Prof. Jens Anders

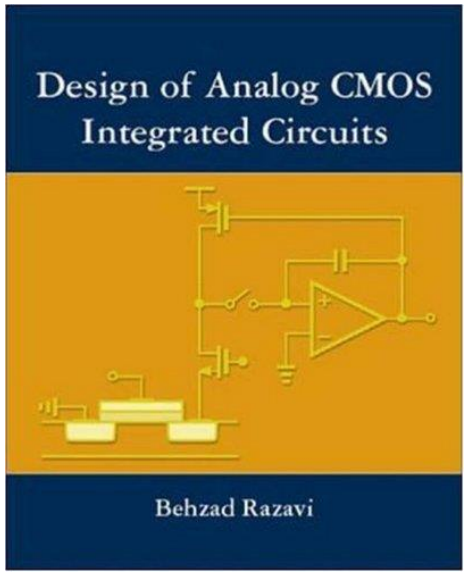
Institute of Smart Sensors
University of Stuttgart

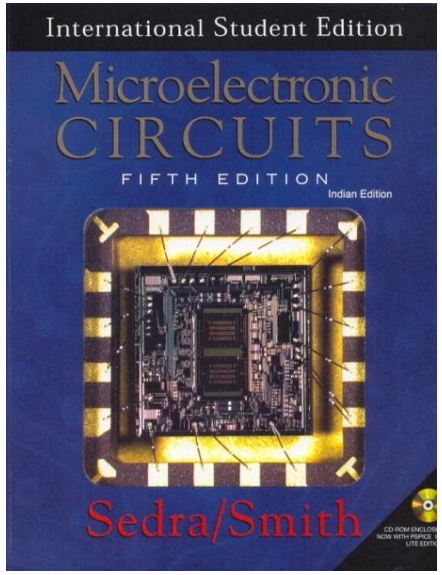
With their permission this course is based on the lectures by Prof. Ortmanns and Kuhl taught in Ulm and Freiburg, respectively.

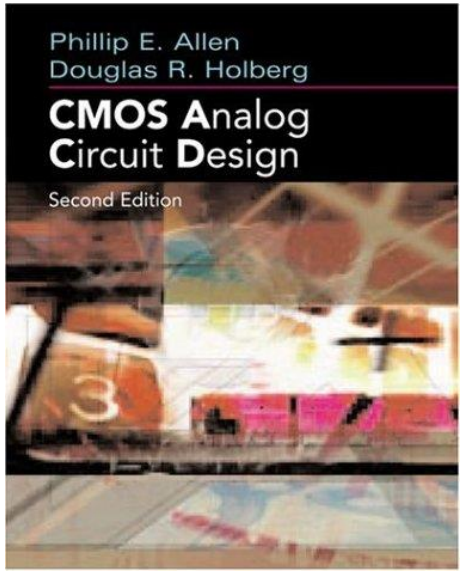


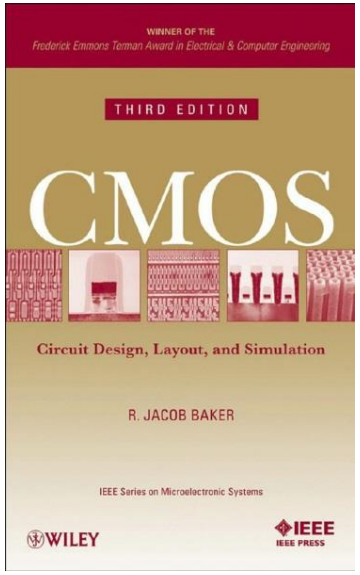


Essential Books









- B. Razavi, *Design of Analog CMOS Integrated Circuits*
- A. S. Sedra and K. C. Smith, *Microelectronic Circuits*
- P. E. Allen and D. R. Hohlberg, *CMOS Analog Circuit Design*
- R. J. Baker, *CMOS Circuit Design, Layout, and Simulation*

This Course

- All information on https://www.iis.uni-stuttgart.de/
  - Teaching
  - Introduction to CMOS Analog Integrated Circuit Design

- Lecture
- Lecture recordings and course material in the Ilias course

- Exercises
- Dr. Nima Shapari

- Exam
- Written or Oral examination (depending on the number of participants, >20 participants ⇒ written)

Microelectronics Course – Contents & Goals

- Basics of analog CMOS circuit design
- Review of CMOS technology and available components
  - Small signal equivalent circuit
  - Current sources
  - Single-stage amplifier and frequency behavior
  - Differential amplifiers
  - Noise in electronic circuits
  - Analog layout
  - A/D & D/A converters
  - MEMS applications
- Goals
  - Understanding of MOS transistor in design and layout
  - Understand to „read“ & design of basic analog integrated circuits
  - Understanding of systems on block and transistor level
  - Understand circuit designers „language“
  - Get fun in creative integrated circuit design

Microelectronics Course – Requirements

- Electrical engineering
  - Ohm’s law and Kirchhoff's node and loop laws “KVL & KCL”

- Electronics
  - Semiconductor basics and physics
  - Semiconductor technology
    - Doping, minority and majority charge carriers (holes and electrons), substrates, masks, lithography, etc…
  - MOS device physics
  - CMOS inverter operation basics

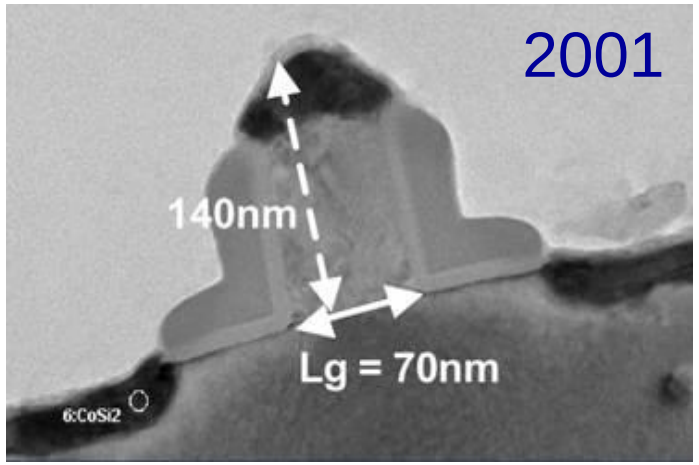
- Signal & system theory
  - Frequency behavior, transfer function, s-syntax, poles & zeros
  - Bode plot, phase margin, open-loop ↔ closed loop

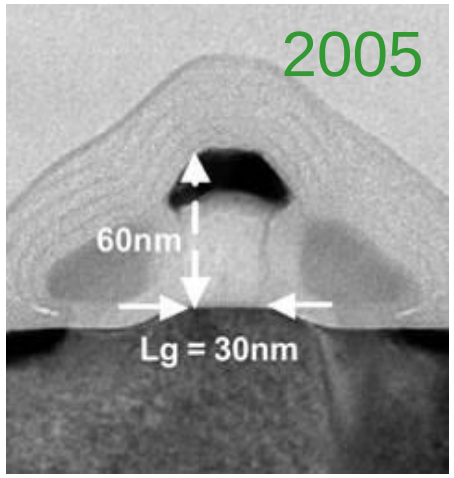
The 1st transistor

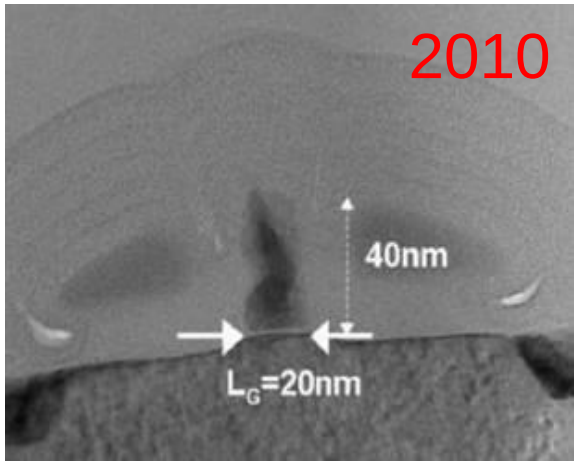


John Bardeen & Walter Brattain Bell Laboratories, 1947

Transistor Scaling

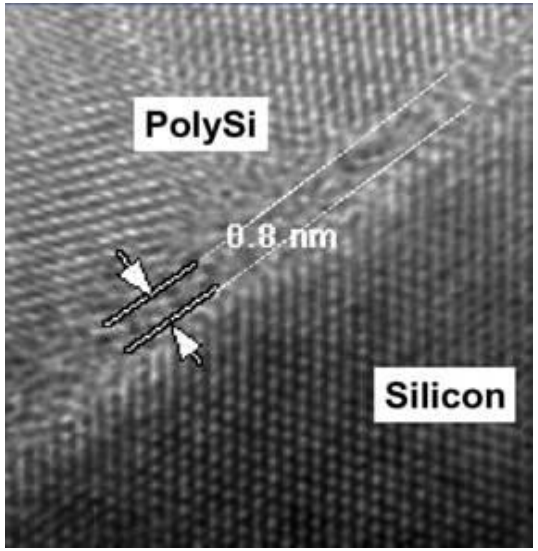


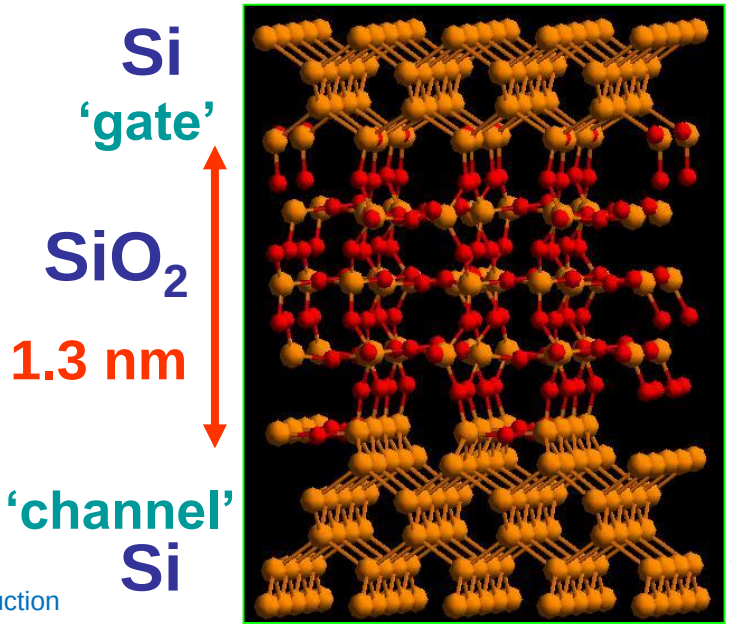




Atomic scale

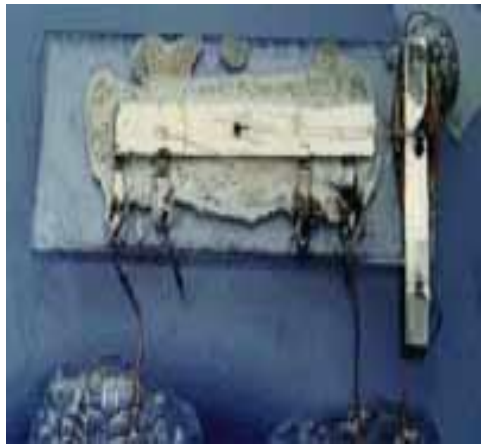
Gate oxide:
3 atomic layers





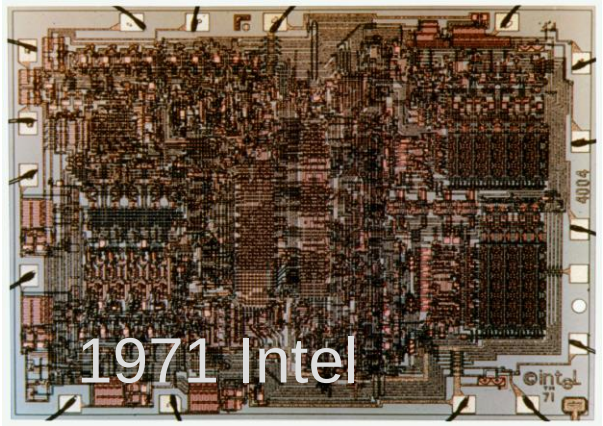
Many transistor realize an Integrated Circuit

First Chip, Two Transistors



1961
Jack Kilby, *TI*
& Robert Noyce,
Fairchild

First Processor, 2300 Trans.

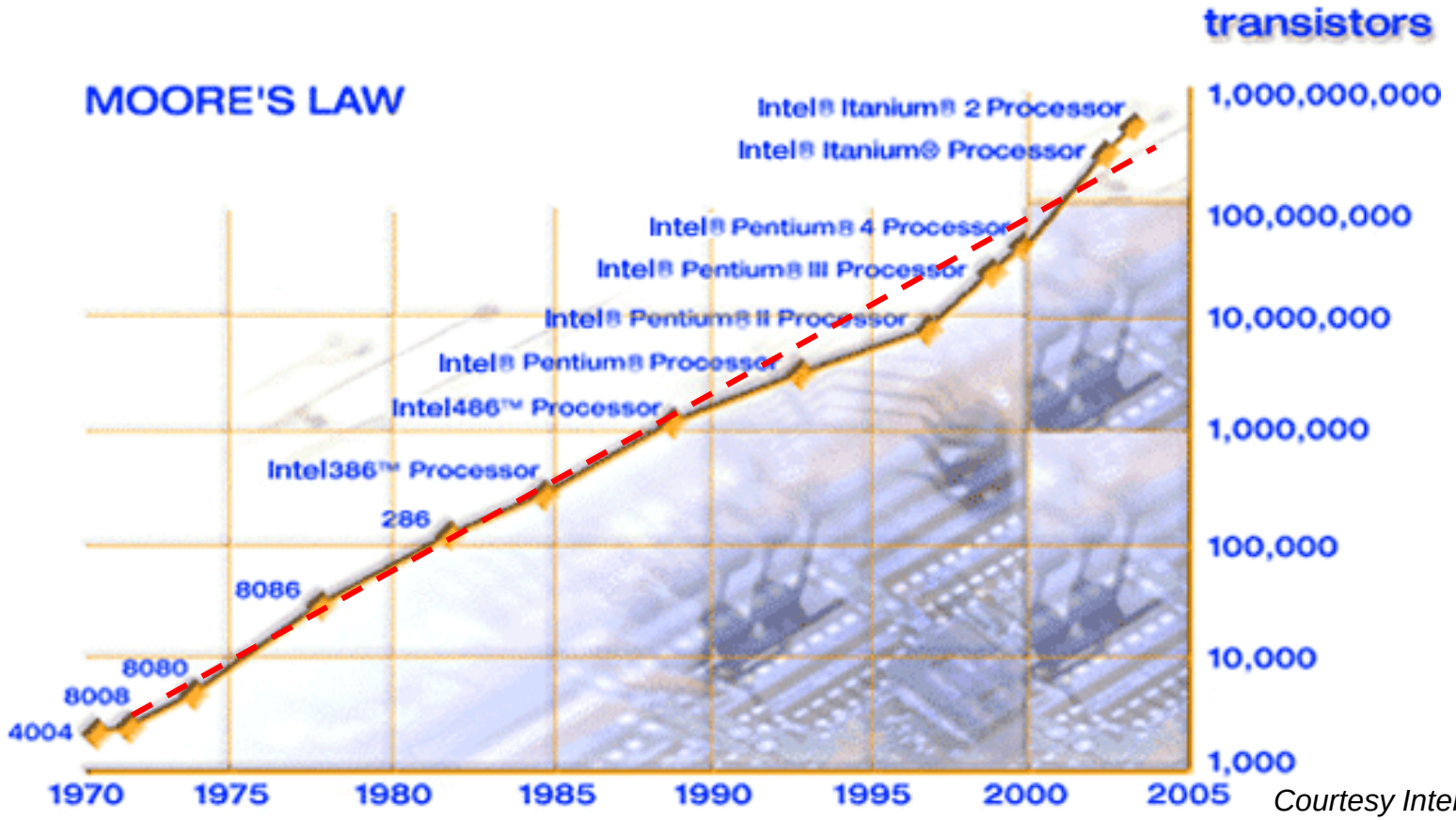




Pentium IV, 42 Million Trans.
2001, Intel

Moore‘s Law

- „The number of transistors per chip will double approximately every 2 years.“ Gordon Moore, Intel 1975.



Courtesy Intel Corp.

- Moore‘s law hold true for digital CMOS circuits over >30y !!!

Moore‘s Law: different view

"If GM had kept up with technology like the computer industry has, we would all be driving $25.00 cars that got 1,000 miles to the gallon".

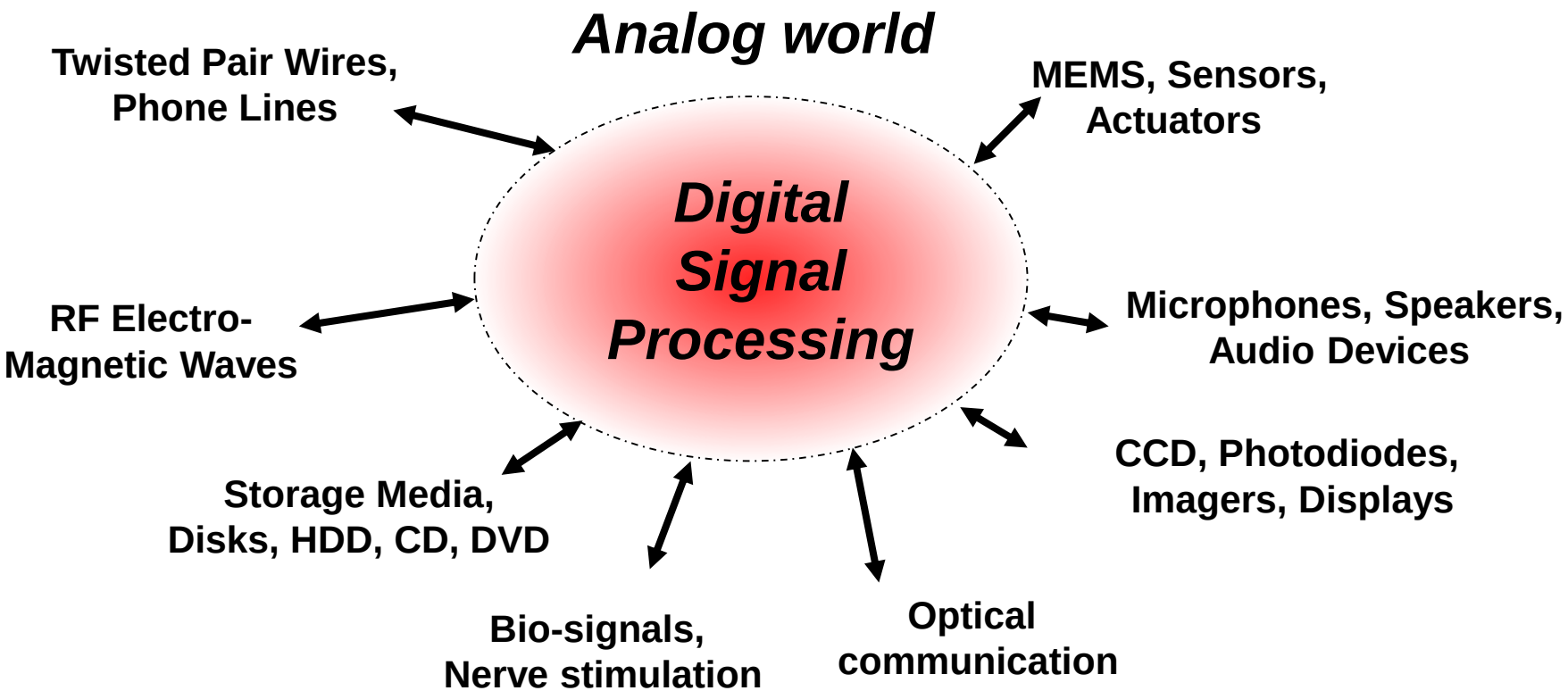
– Bill Gates, *COMDEX* keynote speech

Digital Signal Processing

- Noise immunity, robustness.
- “Unlimited” precision or accuracy.
- Flexibility, programmability, and scalability.
- Electronic design automation (EDA) tools widely available and successful.
- Directly benefiting from **Moore’s law** – “The number of transistors on a chip doubles every 24 months,” *IEDM*, 1975.
- Cost/function drops 29% every year.
- That’s 30x in 10 years.

Necessity for Analog Design

- Remember: Moore‘s law is wonderful for digital design
  - Available functionality increases exponentially
- BUT: real world is analog



Analog Design und Moore‘s Law

Tony Bonaccio von IBM

„This is our future task: the number of analog components should be minimized and be compensated for with the – practically free – digital components. Be it with Auto-calibration or analog self test.“

Design Automation Conference 2004

„Analog can‘t keep up with the scaling. Our designs can‘t always by scaled. ONLY innovations get us further.“

Bo Bitts, Texas Instruments:

„At TI analog and RF-circuits shrink at the same speed as digital circuits. But this is no scaling, but architectural innovation.“

Analog Signal Processing

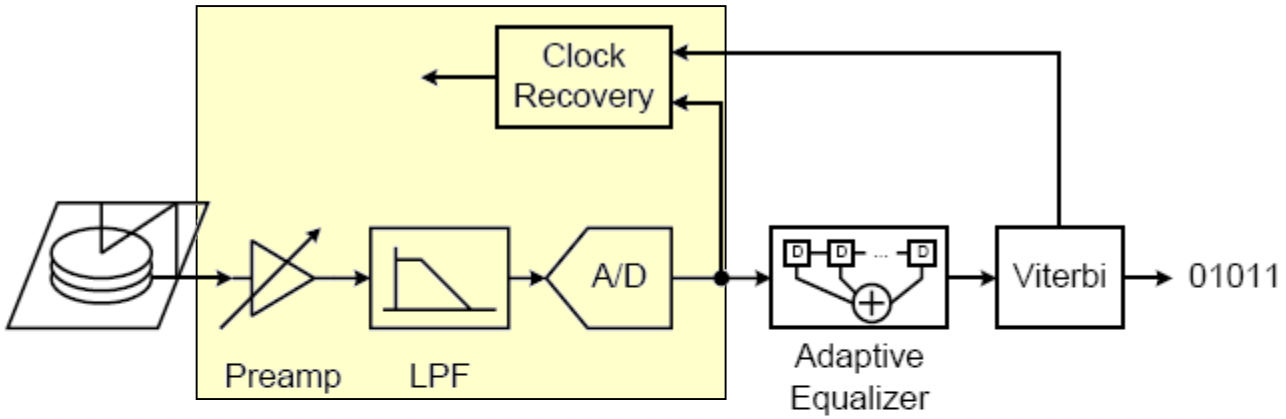
- Sensitive to ***noise*** – SNR (signal-to-noise ratio).
- Subject to device ***non-linearities*** – (harmonic distortion).
- Sensitive to device ***mismatch*** and process variations.
- Difficult to ***design***, simulate, layout, test, and debug.
  - Design automation hardly available
- Inevitable, often limits the ***overall system performance***.
- ***Scaling*** scenario:
  - Enjoyed scaling until ~0.35-µm technology node.
  - High speed circuits keep benefiting (e.g. high-speed ADCs)
  - High SNR design difficult to scale with low supplies (« 3.3V).

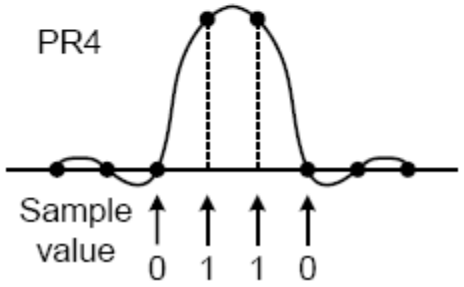
Analog Design → First Conclusion

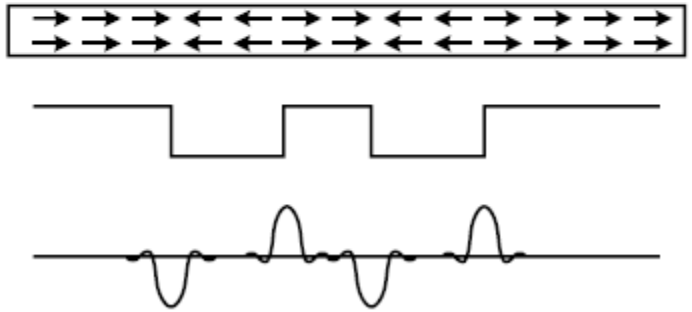
- Trend: SOC 80-90% digital, 10-20% analog
  - CMOS process optimized (& scaled) for digital IC
  - Solve as much in digital as possible
  - DIGITALLY ASSISTED ANALOG INTEGRATED CIRCUITS
- Pass/success ratio :
  - Digital Design ~1 ↔ Analog Design ~ 2 – 3
- Analog design can‘t be automated
  - Tools to facilitate, but not to replace analog designer
  - No standard cell libraries
  - E.g.: OpAmp: thousands/millions of different designs exist over the years. Every application and every process needs some modifications.
- Analog design is ***multifaceted***, ***intuitive*** and always ***new***

Collecting Analog Blocks I

- Example 1: PRML (Partial response maximum likelihood) HDD Read Channel





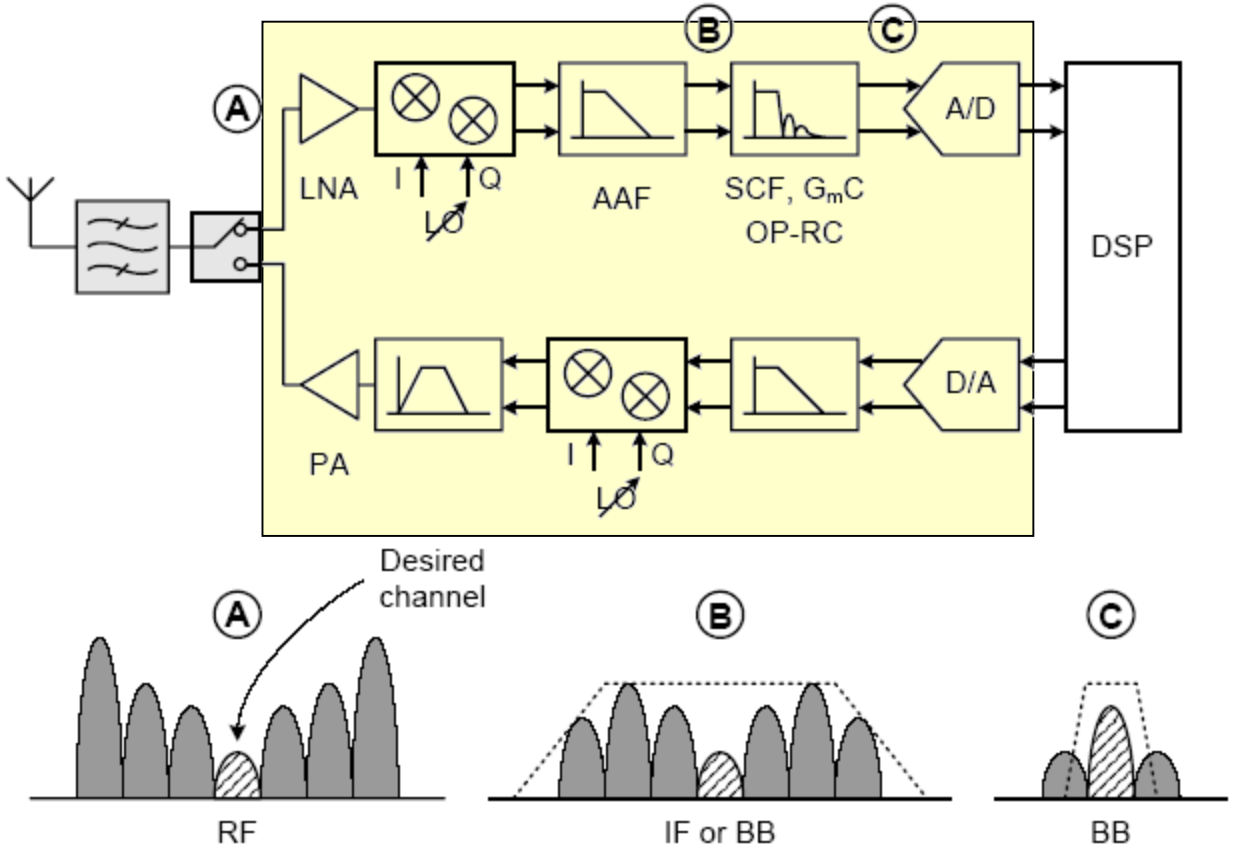


- Filter
- A/D
- Equalizer
- Clock recovery
- Amplifier (LNA)

Courtesy ECE Illinois, Prof. Y. Chiu

Collecting Analog Blocks II

- Example 2: RF Transceiver

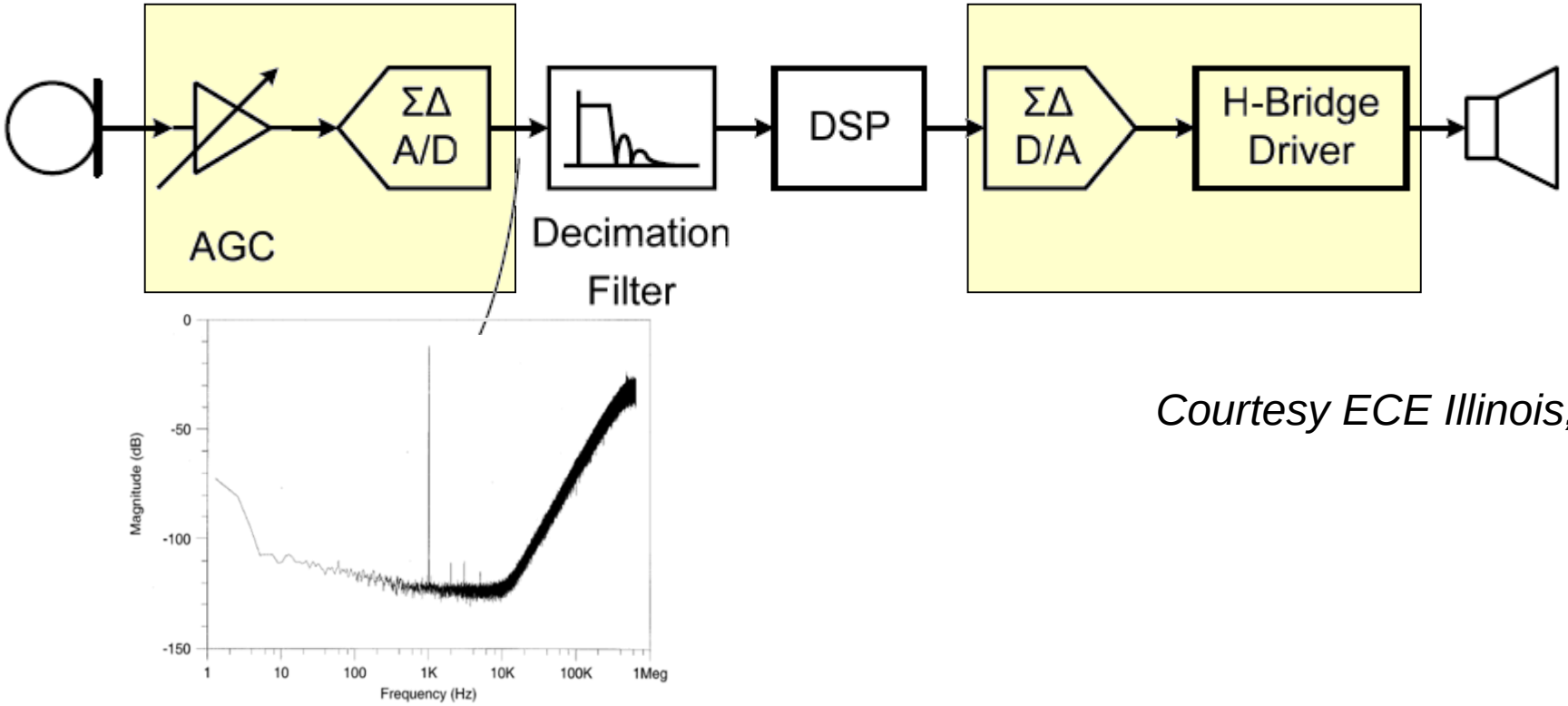


- Filter
- Low Noise Amplifier
- Mixer
- A/D
- D/A
- Power Amplifier

Courtesy ECE Illinois, Prof. Y. Chiu

Collecting Analog Blocks III

- Example 3: Hearing Aid

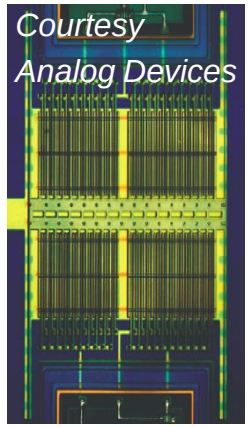


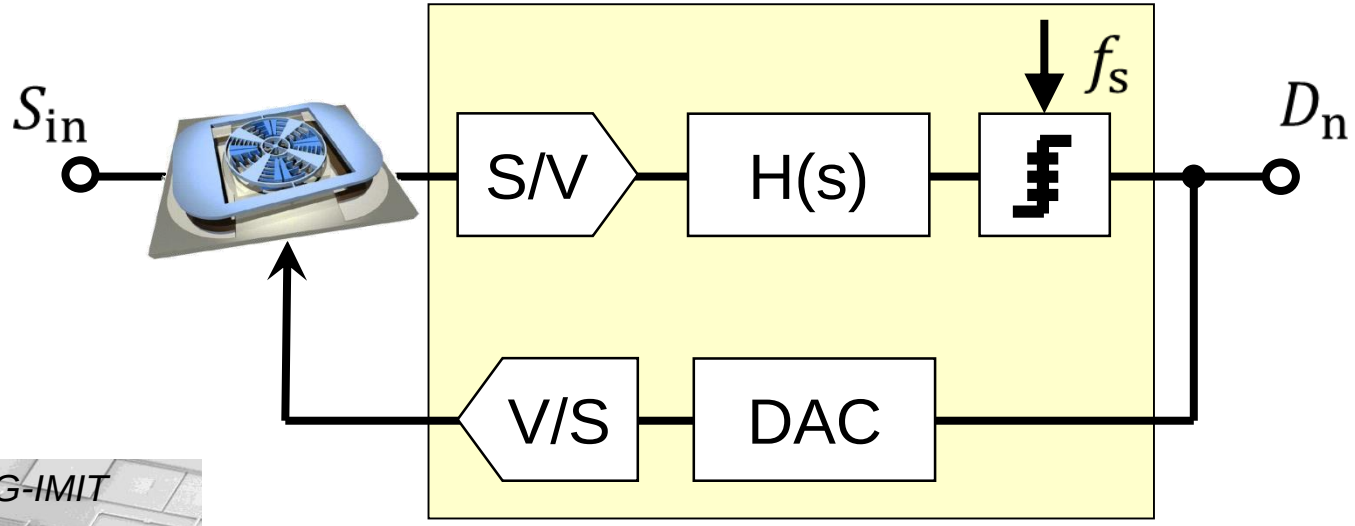
Courtesy ECE Illinois, Prof. Y. Chiu

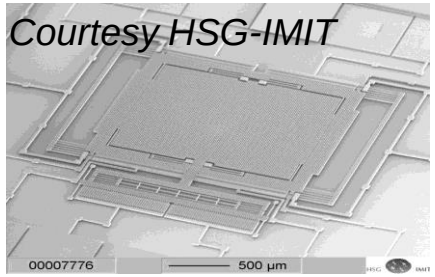
- Automatic Gain Control
- Filter
- ΣΔ A/D
- ΣΔ D/A
- Driver

Collecting Analog Blocks IV

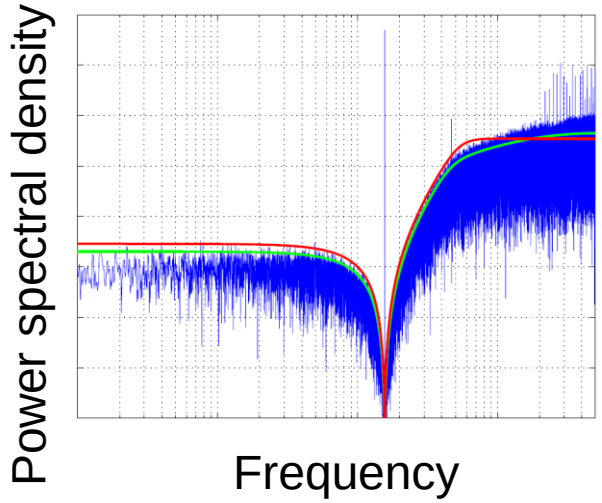
- Example 4: Closed loop ΣΔ sensor readout





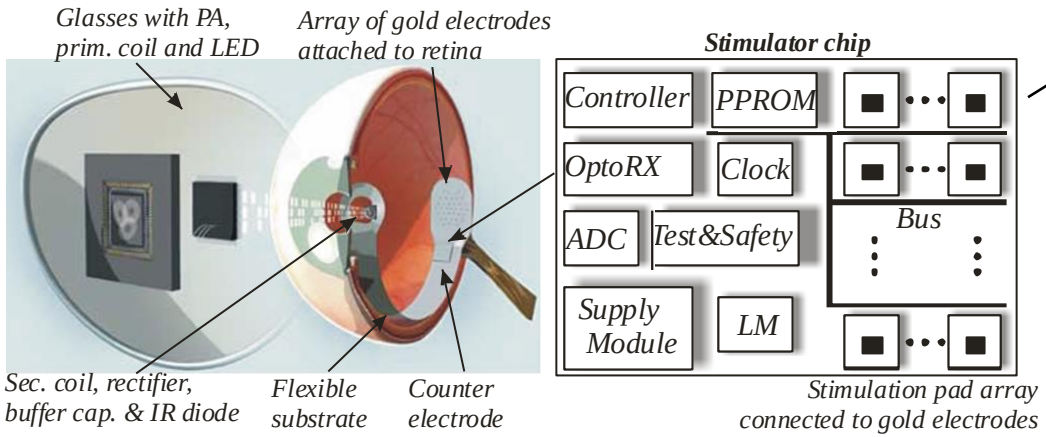


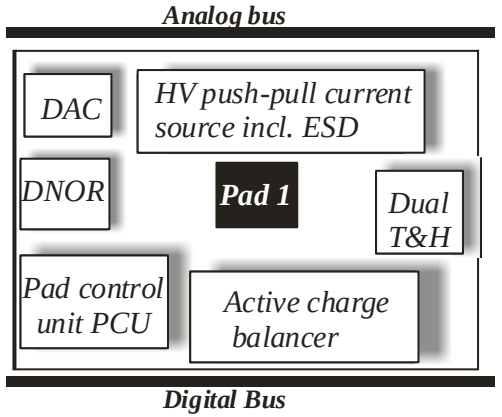
- Charge amplifier
- Loop filter (amplifier)
- D/A
- Comparator , A/D

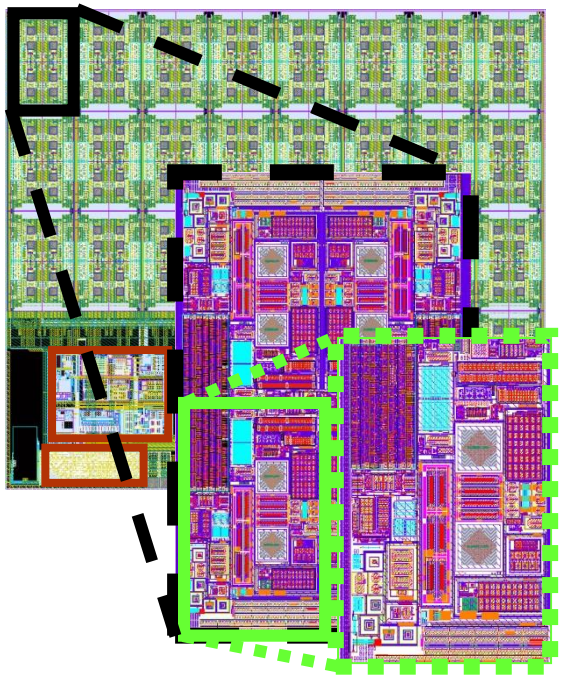


Collecting Analog Blocks/Functions V

- Example 5: Retinal Implant







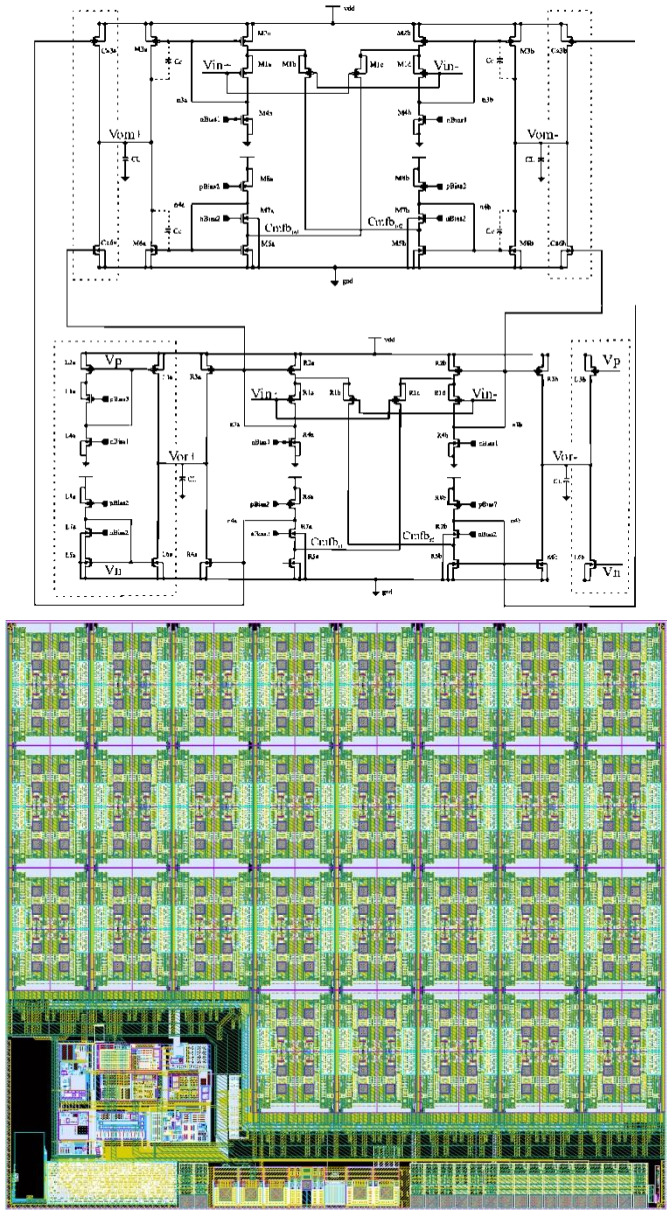
Courtesy sci-worx GmbH & IIP Technologies GmbH

- OptoRX (TIA)
- A/D
- Supply control
- I/O & ESD safety
- Clock recovery
- D/A
- HV driver
- Etc…

Collecting Analog Blocks – Conclusion

- Amplification
- Filtering
- Signal Conditioning (Equalizer)
- Clock and data recovery
- Analog-Digital-Analog Conversion
- Drivers (High current, high voltage)
- Modulation & Demodulation
- Mixing
- ESD and I/O
- Etc…

- <u>INTERFACING PART OF A SYSTEM</u>



Overview

1. Introduction, Motivation and Short Recap
2. Biasing, Single Stage Amplifiers
3. Two stage amplifiers
4. Large Signal Behavior and Distortion
5. SC-Circuits
6. Electronic Noise
7. References / Bandgap Circuits
8. D/A Converters
9. A/D Converters
10. Sigma Delta Modulator
11. Sensor Readout

Circuit examples & system applications

References

- B. Razavi, *Design of Analog CMOS Integrated Circuits*. Boston: McGraw-Hill, 2000
- R. Jacob Baker, *CMOS, Circuit Design, Layout and Simulation*, Wiley-Interscience 2007
- Willy M.C. Sansen, *Analog Design Essentials*, Springer, 2006
- P. E. Allen and D. R. Holberg, *CMOS Analog Circuit Design*, Holt, Rinehart and Winston, 1987.
- P. R. Gray and R. G. Meyer, *Analysis and Design of Analog Integrated Circuits*, John Wiley & Sons, 1984.
- K. R. Laker and W. M. C. Sansen, *Design of Analog Integrated Circuits and Systems*, McGraw Hill, 1994.
- U. Tietze, C. Schenk, Halbleiter Schaltungtechnik, Springer, 1990

Danksagung

Many figures, contents and calculations of this lecture have been prepared at the chair of Microelectronics at the University of Freiburg under
Prof. Yiannos Manoli.

We thank him very much for the possibility of the reuse!

Recap of Stability
of linear time-invariant Systems

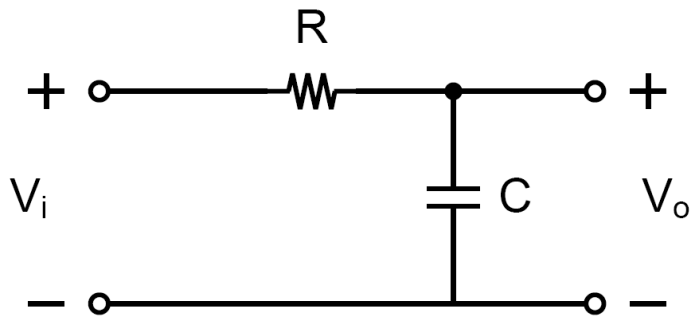
Recap of System Control Theory

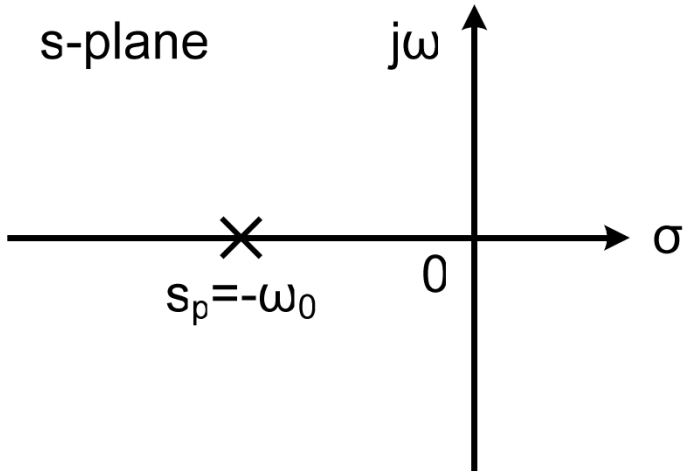
- Transfer function
  - Poles and zeros
- Stability of a system
  - Open loop transfer function
  - *Characteristic equation*
- Methods to prove stability
  - *Solutions of characteristic equation*
  - *Nyquist criterion*
  - *Nyquist plot*
  - Bode plot
  - Phase margin / gain margin

- *See „Systemtheorie“*

Simple RC Filter

- Transfer function and root locus



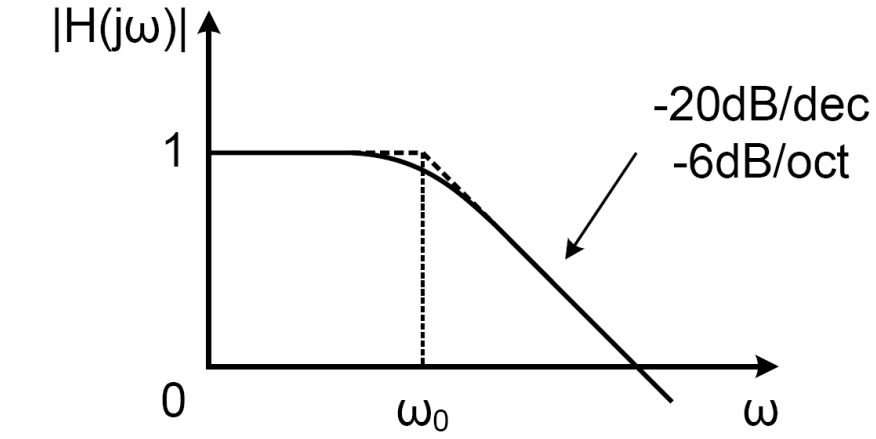


$$H(s) = \frac{V_o}{V_i}(s) = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{1 + sRC}$$

$$s_p = -\omega_0 = -\frac{1}{RC}$$

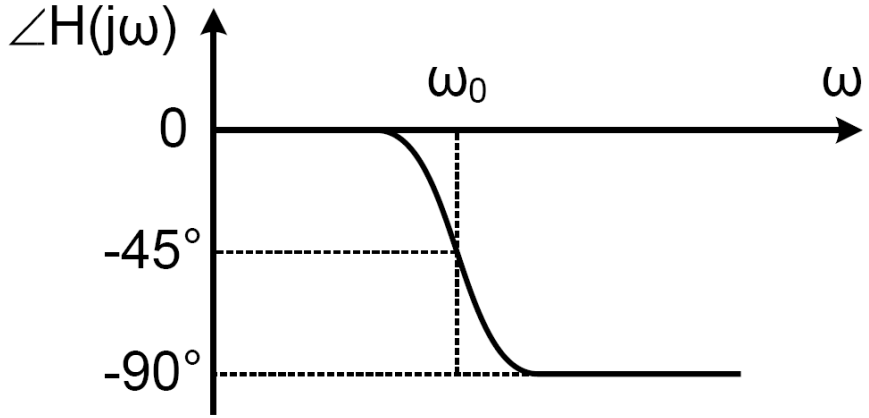
Simple RC Filter

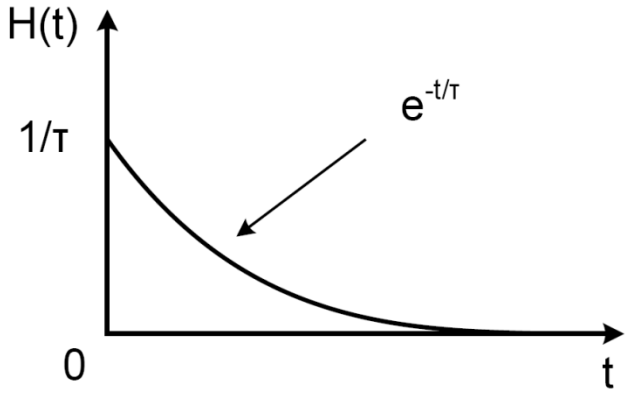
- Bode plot and step response



$$h(t) = \frac{1}{\tau} e^{-\frac{t}{\tau}} \quad for \quad t \geq 0,$$

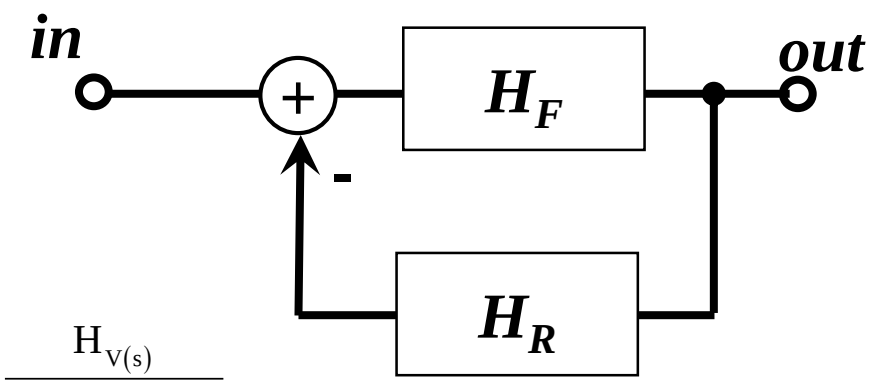
$$\tau = RC, \quad \omega_0 = \frac{1}{RC}$$





Stability

- Non-stable circuits need
  - More than one pole
  - Gain
  - e.g. Oscillator, feedback systems



- Transfer function

$$\frac{\text{out}}{\text{in}} = H_{(s)} = \frac{H_{V(s)}}{1 + H_{R(s)} H_{V(s)}}$$

- Characteristic equation

$$1 + H_{R(s)} H_{V(s)} = 1 + H_{O(s)} = 0$$

- With s_i being solutions

$$a_n (s - s_1)(s - s_2)\ldots(s - s_n) = 0$$

- These solutions determine the intrinsic response

$$\text{out}_h = C_1 e^{s_1 t} + C_2 e^{s_2 t} + \ldots + C_n e^{s_n t}$$

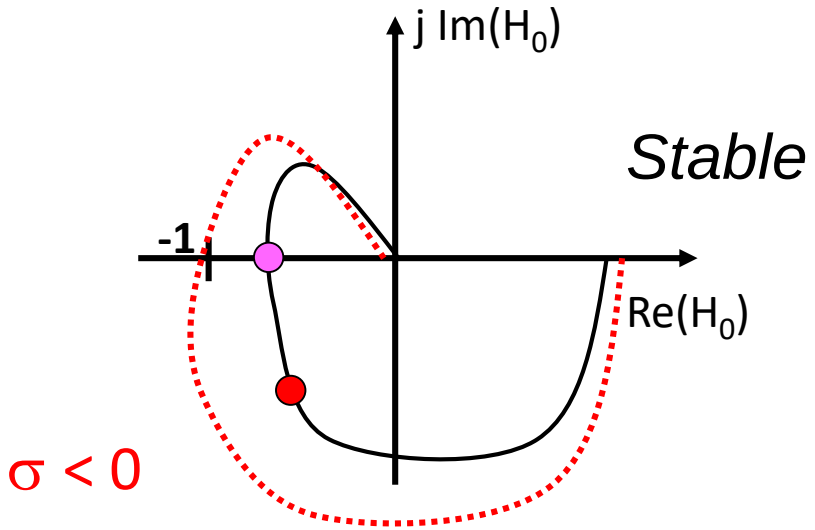
$$s = \sigma + j\omega$$

- Remember

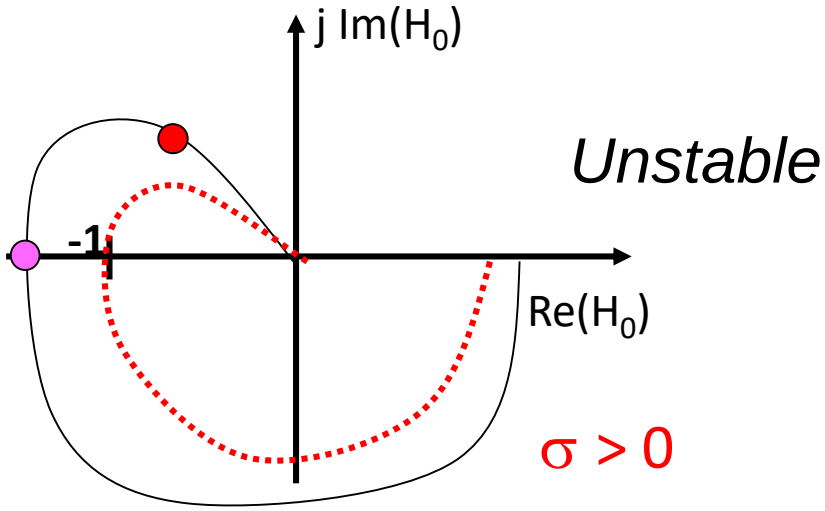
$$\sigma < 0 \rightarrow \text{Stable partial response}$$

Stability – Criterion I

- **Nyquist Plot**
  - Complex graphical representation of $H_0(j\omega)$
  - Solution of characteristic equation $H_0(s)=-1$ with $\sigma>0$ or $\sigma < 0$??



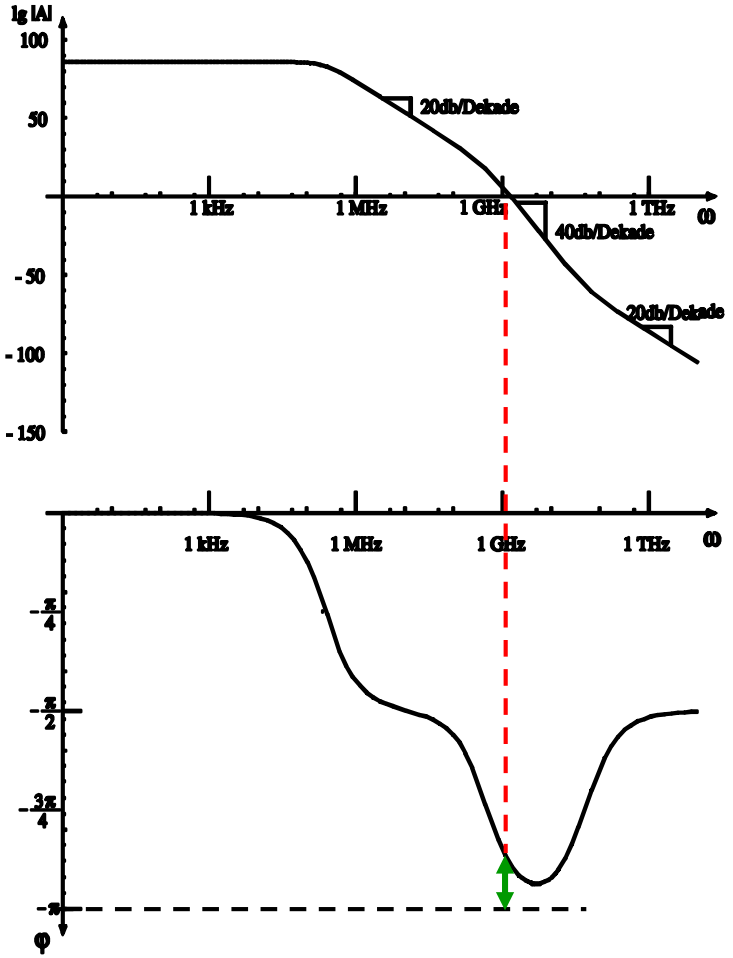
- Stability: means for $\sigma=0$
  - If we cross 180° phase shift → amplitude must be <1
  - If amplitude ==1 → phase shift must not be >180°



e.g. http://www.facstaff.bucknell.edu/mastascu/eControlHTML/Freq/Nyquist3.html

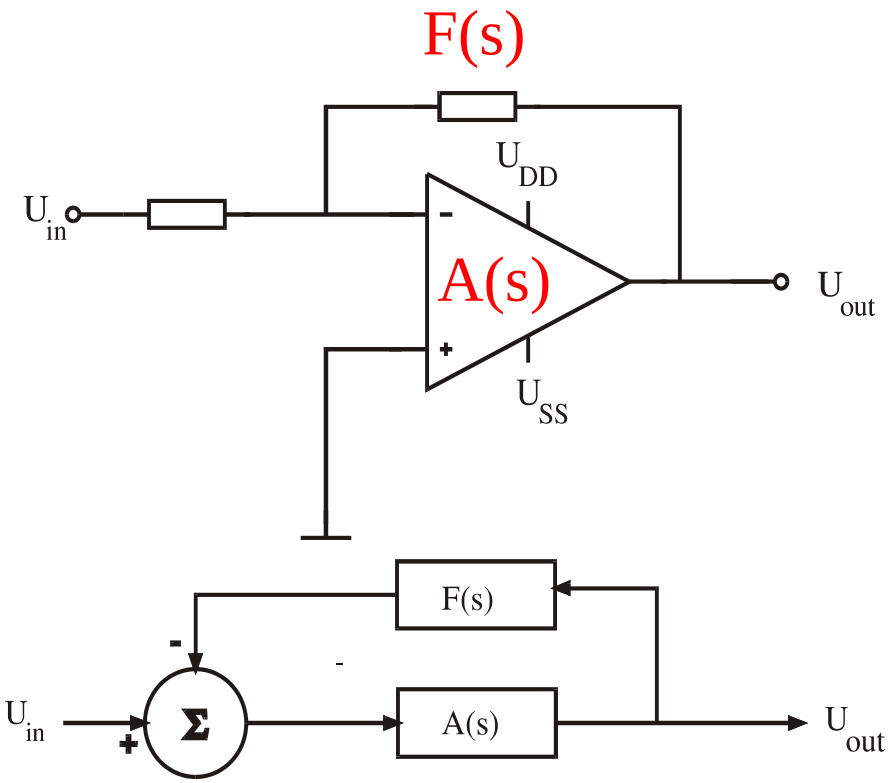
Stability – Criterion

- **Bode Plot**
  - Graphical representation of $H_0(j\omega)$
  - Split into phase and magnitude
- Nyquist stability translates to:
  - **Phase margin**
    If amplitude ==1
    → what phase difference do we have to -180°
  - Gain margin
    If we cross 180° phase shift
    → how large is the damping corresponding to $|H_0(j\omega)|=1$
    ($|H_0(j\omega)|$ =0dB)



stability_CMOS06.m

Negative Feedback



$$U_{out} = A(s) \cdot \left(U_{in} - U_{out} \cdot F(s) \right)$$

$$\frac{U_{out}}{U_{in}} = \frac{A(s)}{1 + A(s) \cdot F(s)} = \frac{1}{\frac{1}{A(s)} + F(s)}$$

$$\approx \frac{1}{F(s)}$$

Function of system only dependend on feedback circuit:

if A(s) >>

Stabilität

$$@ \; s \; \left| \; \varphi = 180° \right. \quad \Rightarrow \left| A(s) \cdot F(s) \right| < 1$$

Gain Margin

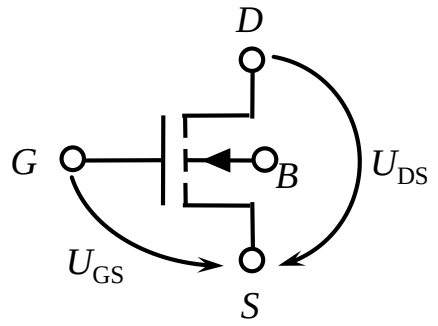
$$\text{oder} \quad @ \left| A(s) \cdot F(s) \right| = 1 \quad \Rightarrow \varphi_{res} = 180° - \varphi > 0$$

Phase Margin

Recap of CMOS Circuits

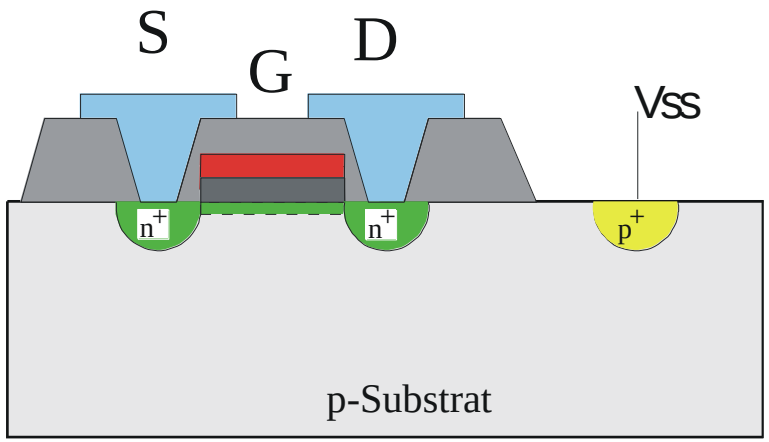
Our Active Device – MOS Transistor

- Is a voltage controlled current source



- Right potential between gate and bulk
  - Makes inversion layer → channel
  - Channel between Drain and Source
  - Current can flow
  - $V_{GS}>V_{TH}$

- Too „much“ potential V_{DS}
  - Channel is cut off
  - Channel length is “modulated”



- If V_B is not equal to V_S, threshold voltage V_{th} changes

Output I_D vs V_{DS} and Some Equations

- Saturation

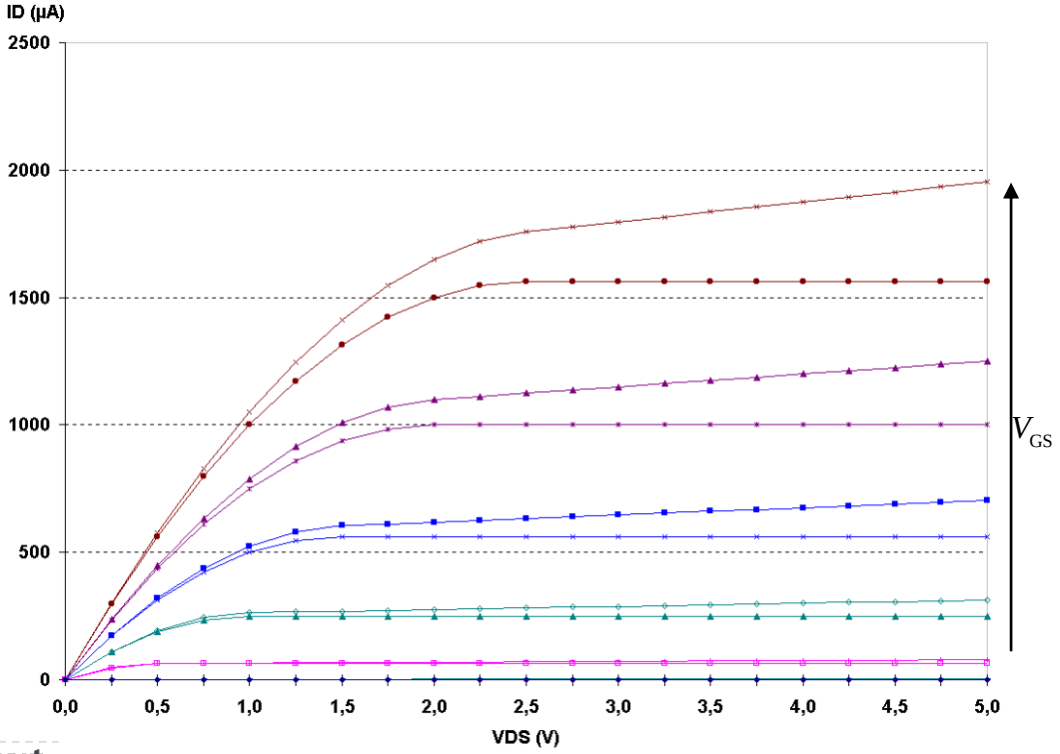
$$I_D = \frac{\beta_0}{2}\frac{W}{L}\cdot\left(V_{GS}-V_T\right)^2\cdot\left(1+\lambda\cdot V_{DS}\right)$$

- Linear operation

$$I_D = \beta_0\frac{W}{L}\cdot\left(V_{GS}-\frac{V_{DS}}{2}-U_T\right)\cdot\left(1+\lambda\cdot V_{DS}\right)$$

- Subthreshold

$$I_D = K\cdot e^{\frac{V_{GS}-V_t}{S\cdot\ln 10}}\cdot\left(1-e^{\frac{-qV_{DS}}{k\cdot T}}\right)$$



Spice Level 1 vs. Level 13 BSIM

- Spice level 1
  - Channel length modulation
  - Substrate steering of $V_{th}=f(V_{bs})$

- What to do
  - Only simulate ?
  - Only calculate ?
  - Both ?

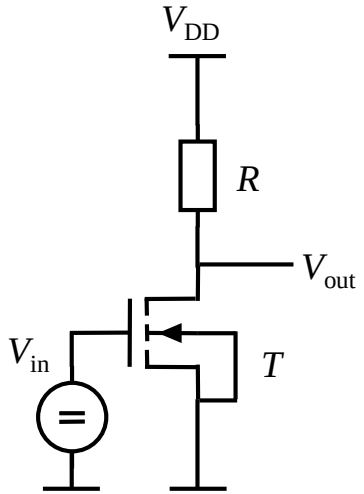
- BSIM
  - Efficient empirical model
  - Vertical field dependence of carrier mobility
  - Carrier velocity saturation
  - Drain-induced barrier lowering
  - Depletion charge sharing by source and drain
  - Non-uniform doping profile for ion-implanted devices
  - Channel length modulation
  - Subthreshold conduction
  - Geometric dependence of electrical parameters

Click for different models examples

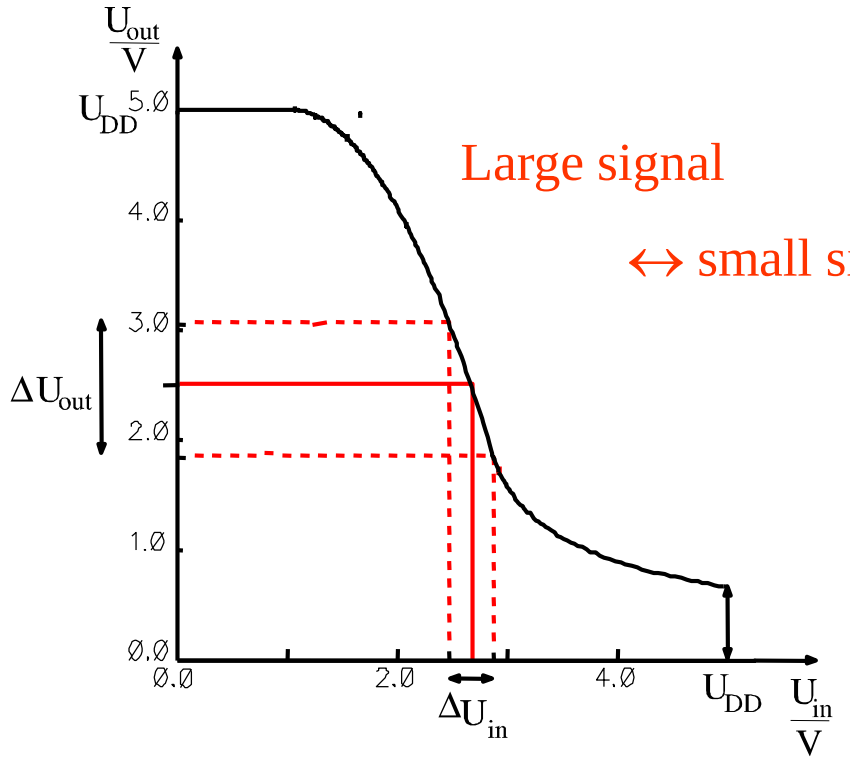
http://www.ece.uci.edu/eceware/hspice_docs/hspice_and_qrg/hspice_2001_2-153.html

Simple MOS Amplifier

- Transistor in its dc operation (bias) point
  - E.g. V_{gs}=2.5V & V_{ds}=2.5V



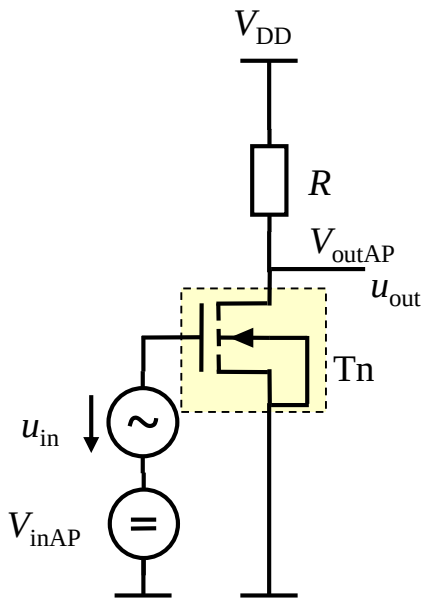


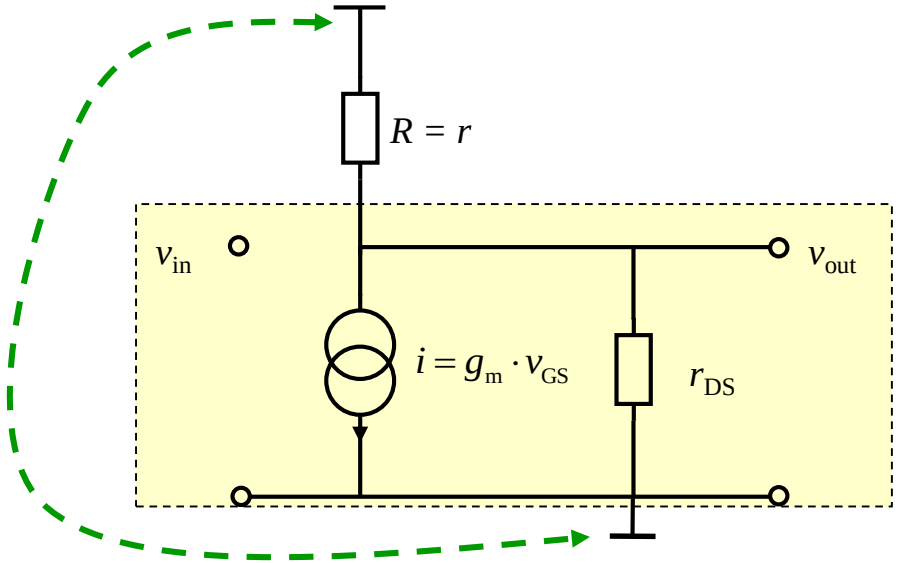


Large signal

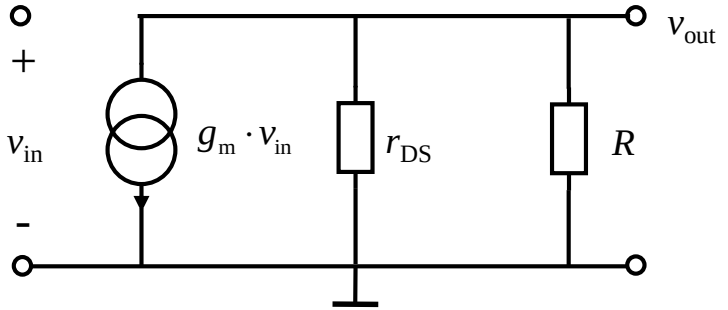
↔ small signal

Simple MOS Amplifier – Small Signal





$\cong$

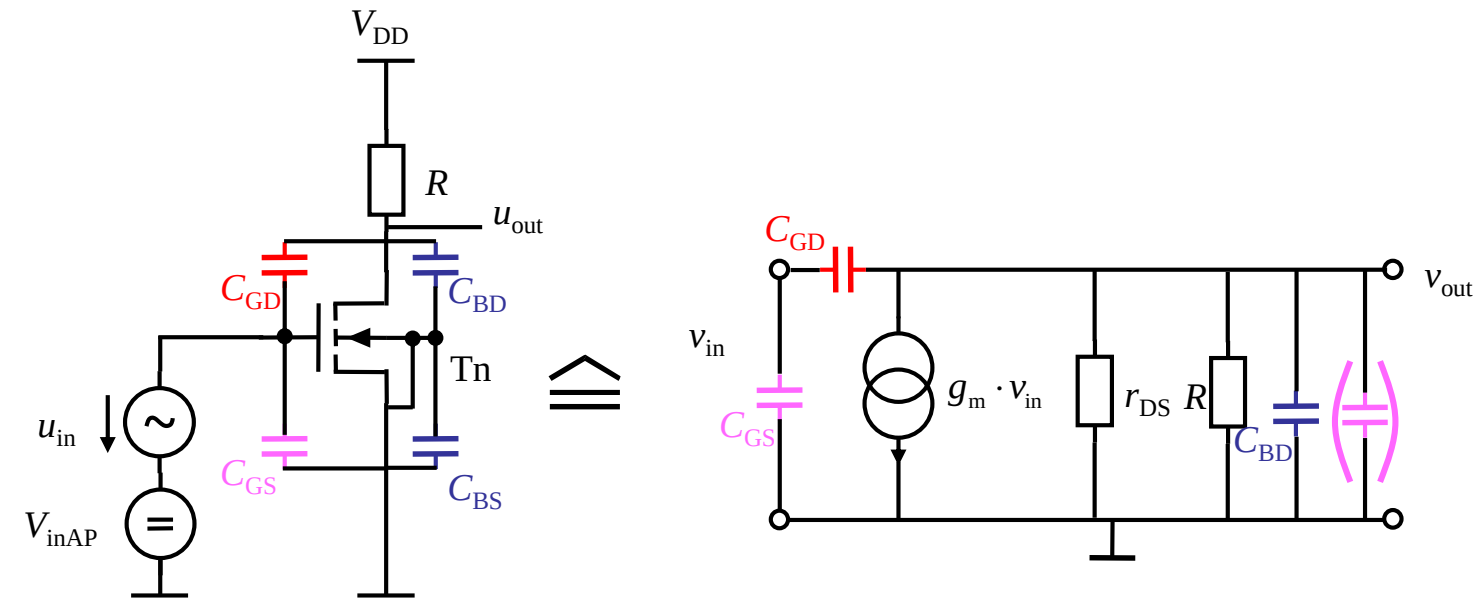


$$A = \frac{\Delta V_{out}}{\Delta V_{in}} = \frac{v_{out}}{v_{in}} = -g_m \cdot r_{out}$$

$$g_m = \frac{dI_D}{dV_{GS}} \qquad r_{out} = \left(R \| r_{DS}\right)$$

$$\boxed{A = -g_m \cdot \left(R \| r_{DS}\right)}$$

Simple MOS Amplifier – Small Signal with Caps



$C_{GS|next}$

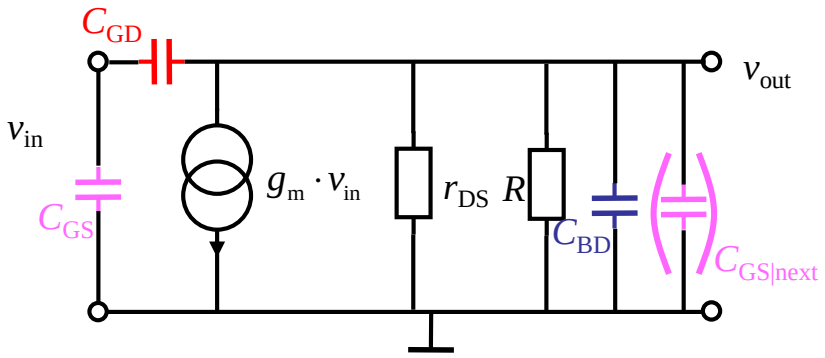
$C_{GS} \rightarrow$ Oxide

$C_{BD} \rightarrow$ pn-Junction

$C_{GD} \rightarrow$ Overlap

$C_{GD} < C_{BD} < C_{GS}$

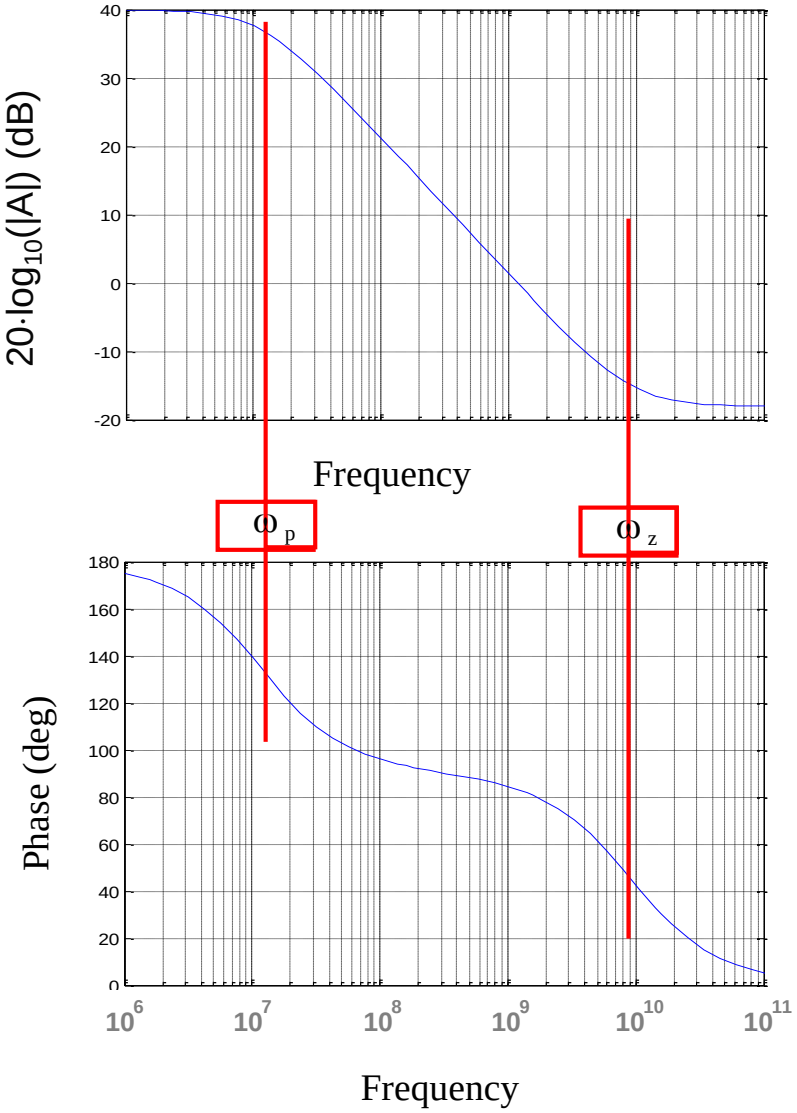
Simple MOS Amplifier – Transfer Function



$$A_V = \frac{v_{out}}{v_{in}} = -(g_m - sC_{GD}) \cdot \left(\frac{r_{DS}}{1 + s(C_{GD} + C_{DB}) \cdot r_{DS}} \right)$$

Pole $$\Rightarrow \omega_p = -\frac{1}{r_{DS} \cdot (C_{GD} + C_{DB})}$$

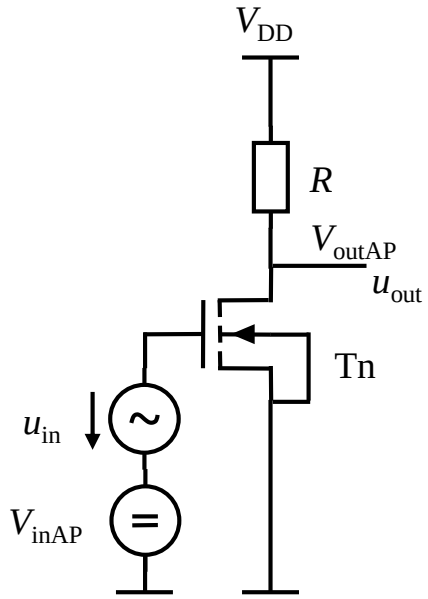
Zero $$\Rightarrow \omega_z = \frac{g_m}{C_{GD}}$$

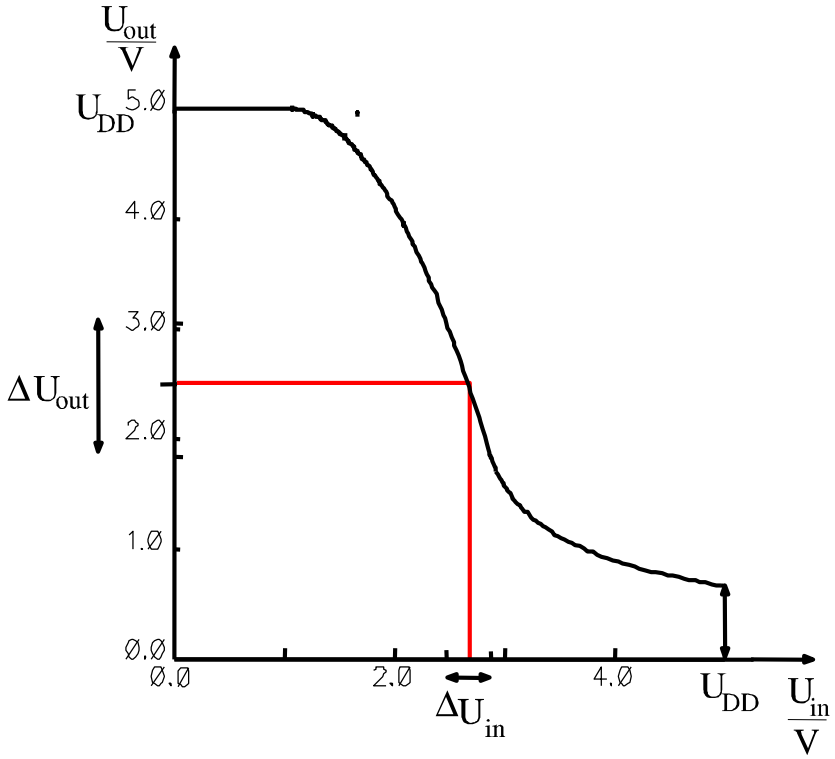


- NOTE:
  - C_{gd} zero causes amplitude to raise but also negative phase shift
  - Miller effect for C_{gd} !!!

Differential Stage – Motivation

- *Biasing of transistor amplifier (common source)*
  - *Slight change alters everything*
  - *W/L, I_b or $V_{gs|b}$ or*



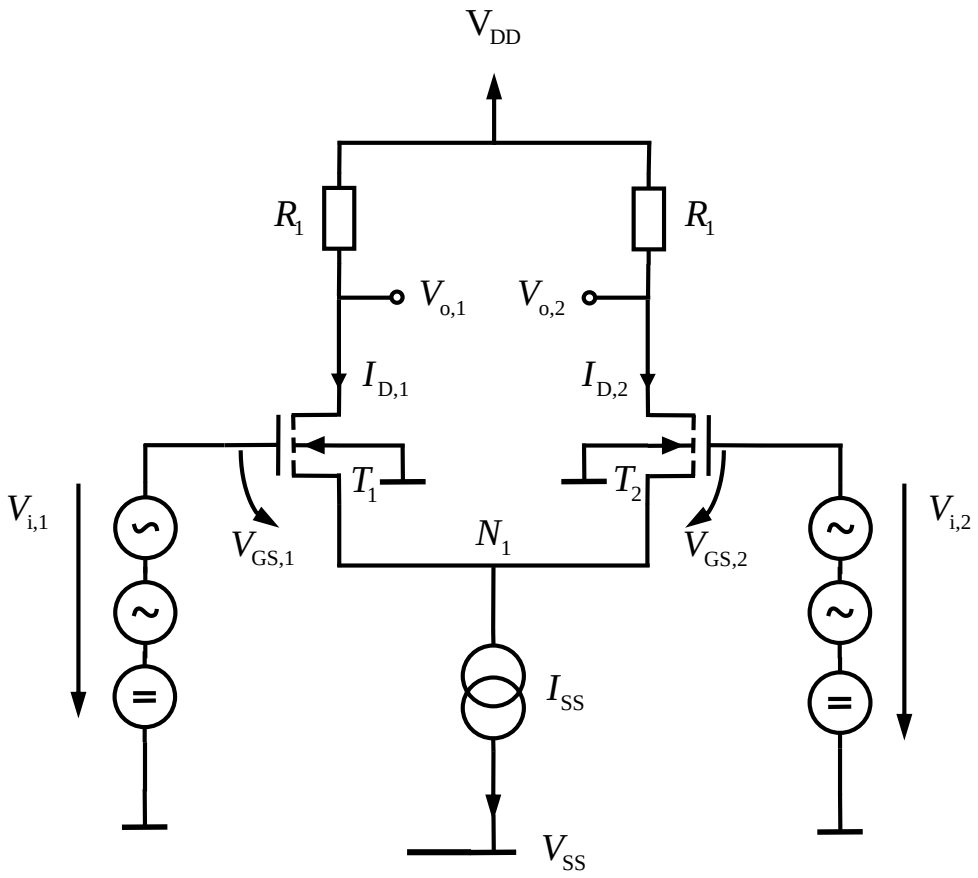


- *Adjustable transfer characteristic*

  $\rightarrow$ *differential amplifier*

Simple MOS Differential Pair – Voltages

- ΔV_i -> ΔI_{out} -> ΔV_o
- Input voltage
  - Biasvoltage
  - Differential signal
  - Common mode signal
- Large ↔ small signal



$$I_{D,1} + I_{D,2} = I_{SS}$$

Common mode: $I_{D,1} = I_{D,2} = \frac{I_{SS}}{2}$

Differential mode: $I_{D,1} = \frac{I_{SS}}{2} + \Delta I$ $\quad I_{D,2} = \frac{I_{SS}}{2} - \Delta I$

Simple MOS Differential Pair – Gain

- Differential gain
  - N_1 stays constant
  - A_{DM} ~ gain of MOS inverter

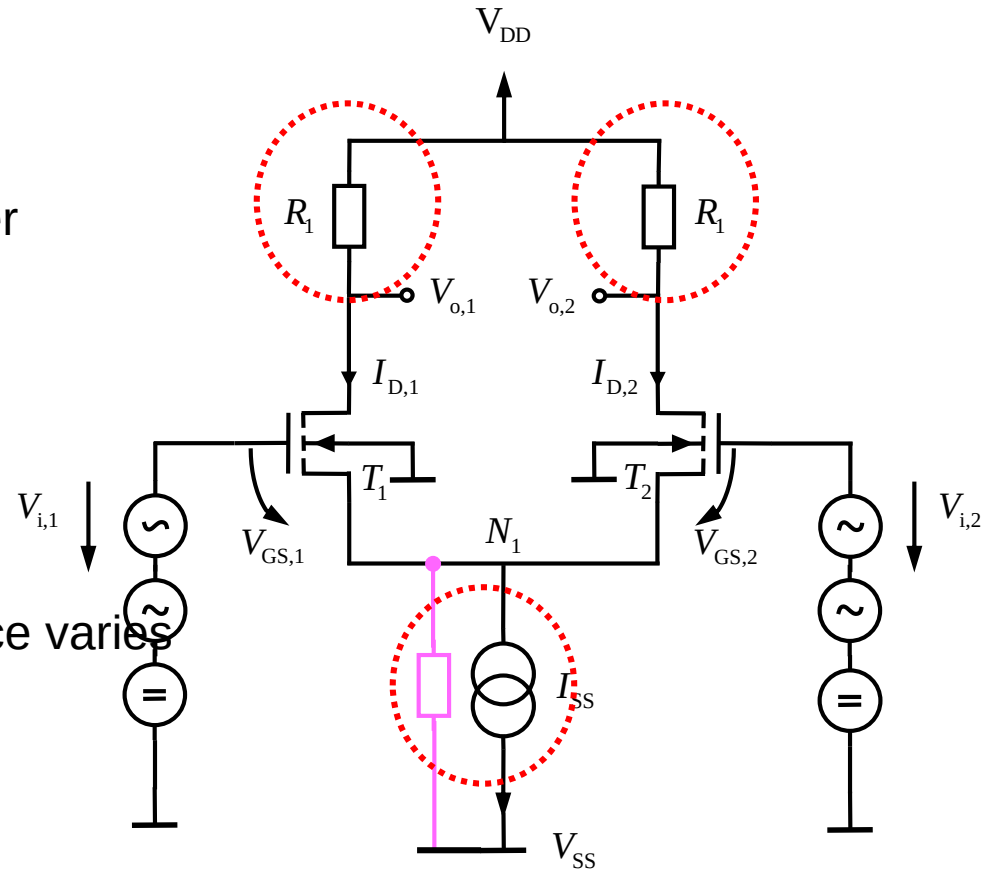
$$A_{vDM} \approx -g_m \cdot r_{out}$$

- Common mode gain
  - N_1 moves
  - Voltage over current source varies
  - Ideal I_{SS} stays constant

$$I_{D,1} = I_{D,2} = \frac{I_{SS}}{2}$$

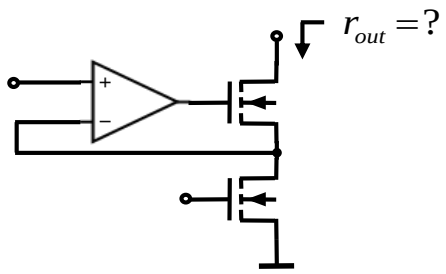
- Non-ideal current source varies

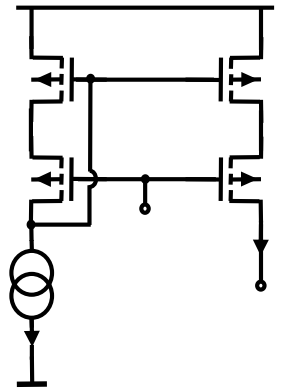
$$A_{vCM} \approx -\frac{R_1}{2 R_{SS}}$$



Outlook: Next Time in This Theatre

- Current sources
- Cascode current sources
- Gain boosting in cascodes
- High swing cascodes





- Single stage differential amplifier
  - Differential pair with active load
  - Telescopic differential amplifier
  - Folded cascode differential amplifier

